Math 125

Jubilee 1

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Instructions:

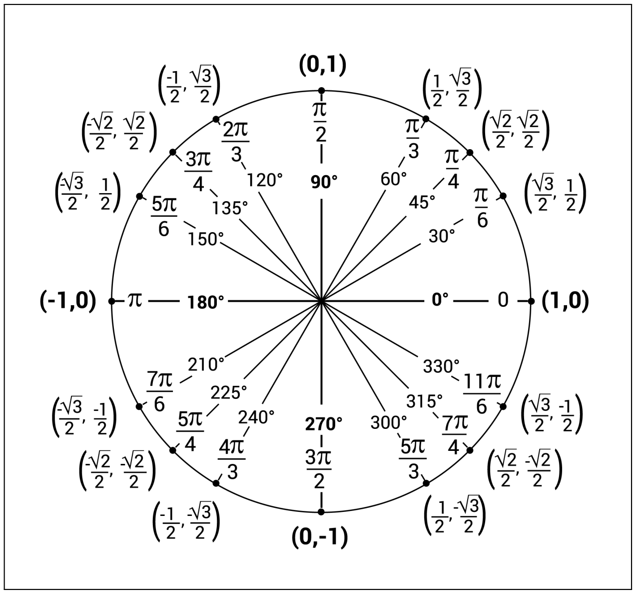
- Make sure to clearly show each step of your solution for full credit - correct answers without any work will not receive credit. Some questions may remind you to show your work explicitly, but you should assume each question requires you to justify your steps by making your thought process clear in your solution. You can use the solutions we write in class as an indication of how much work you are expected to show. You can ask the instructor if it is not clear how much work you are expected to show on a particular problem.
- Write all of your work and answers on this document. If you need extra paper, please ask the instructor. If a box is provided for your final answer, please write your final answer in the box.
- Indicate your affirmation of the following statement by signing below.

I affirm that I did not give or receive any unauthorized assistance to solve the problems below. I understand that the penalty for academic dishonesty is an F for the course and possible suspension or expulsion from the university.

Signature: ______________________________

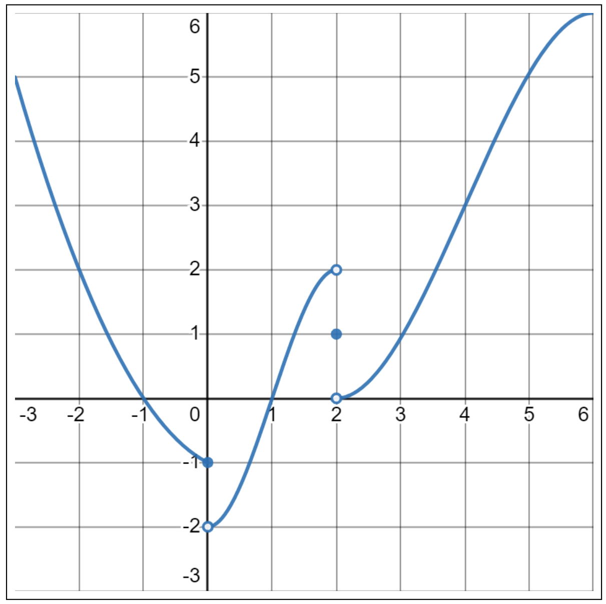
Formulas

- $\lim_{\theta\to 0}\frac{\sin\theta}{\theta} = 1,\ \lim_{\theta\to 0}\frac{\cos\theta - 1}{\theta} = 0$
- Continuity at $x = a$: $\lim_{x\to a} f(x) = f(a)$
- $f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$
- Basic derivatives
  - $\frac{d}{dx}(c\cdot f(x)) = c\cdot f'(x)$
  - $\frac{d}{dx}(f(x)\pm g(x)) = f'(x)\pm g'(x)$
  - $\frac{d}{dx}(x^n) = nx^{n-1}$
  - $\frac{d}{dx}(e^x) = e^x$
  - $\frac{d}{dx}(\ln(x)) = \frac{1}{x}$
  - $\frac{d}{dx}(\sin x) = \cos x$
  - $\frac{d}{dx}(\cos x) = -\sin x$
  - $\frac{d}{dx}(\tan x) = \sec^2 x$
  - $\frac{d}{dx}(\sec x) = \sec x\cdot\tan x$
  - $\frac{d}{dx}(\cot x) = -\csc^2 x$
  - $\frac{d}{dx}(\csc x) = -\csc x\cdot\cot x$
- Derivative rules
  - $\frac{d}{dx}(f(x)\cdot g(x)) = f'(x)\cdot g(x) + g'(x)\cdot f(x)$
  - $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)\cdot f'(x) - f(x)\cdot g'(x)}{[g(x)]^2}$
  - $\frac{d}{dx}(f(g(x))) = f'(g(x))\cdot g'(x)$
- $L(x) = f'(a)(x-a) + f(a)$



A1 - Limits and continuity (inspired by Brightspace problems 1, 5, 6, and 9 and part 3 of the class notes)

(a) For the function g whose graph is given to the right, state the value of each quantity, if it exists. If it does not exist, write DNE. No explanations are necessary for this problem.



i. $\lim_{t\to 0^-} g(t) =$ -1

ii. $\lim_{t\to 0} g(t) =$ DNE

iii. $\lim_{t\to 2^+} g(t) =$ 0

iv. $\lim_{t\to 4} g(t) =$ 3

(b) Find the limits. Clearly show each step of your solution for full credit. If the limit does not exist, explain why.

i. $\lim_{x\to 0} \dfrac{\sin^2(x)}{x\tan(x)}$

$$= \lim_{x\to 0} \frac{\sin(x)}{x} \cdot \frac{\sin(x)}{\frac{\sin x}{\cos x}}$$

$$= \lim_{x\to 0} \frac{\sin x}{x} \cdot \cos x = 1 \cdot 1 = 1$$

ii. $\lim_{x\to -\infty} \dfrac{\sqrt{-x^3+x^2-x+1}}{10x-3}$

$$= \lim_{x\to\infty} \frac{\sqrt{x^3+x^2+x+1}}{-10x-3}$$

$$= \lim_{x\to\infty} \frac{\sqrt{x+1+\frac{1}{x}+\frac{1}{x^2}}}{-10-\frac{3}{x}} = -\infty$$

(c) Find the value of a for which the function $f(x)$ given on the right is continuous at $x = 1$.

$$f(x) = \begin{cases} x^2 - a & \text{if } x \le 1 \\ \dfrac{x^2-1}{x^2+2x-3} & \text{if } x > 1 \end{cases}$$

$$f(1) = 1^2 - a = 1 - a$$

$$\lim_{x\to 1^+} \frac{(x-1)(x+1)}{(x-1)(x+3)} = \frac{1+1}{1+3} = \frac{1}{2}$$

$$1 - a = \frac{1}{2} \rightarrow a = \frac{1}{2}$$

A2 - **The derivative** (inspired by part 2 of the class notes)

(a) Use the limit definition of the derivative to show that if $f(x) = mx + b$, then $f'(x) = m$.

$$\lim_{h\to 0} \frac{m(x+h)+b-(mx+b)}{h} = \lim_{h\to 0} \frac{mx+mh+b-mx-b}{h}$$

$$= \lim_{h\to 0} \frac{mh}{h} = m$$

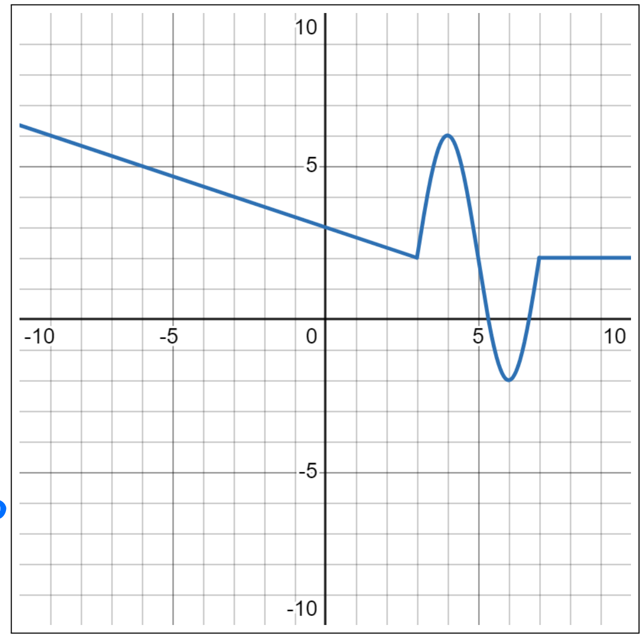
(b) Let $g(x) = \dfrac{k}{x^2}$ for some constant k. Use the limit definition of the derivative to find $g'(x)$.

$$\lim_{h\to 0} \frac{\frac{k}{(x+h)^2} - \frac{k}{x^2}}{h} \cdot \frac{x^2(x+h)^2}{x^2(x+h)^2} = \lim_{h\to 0} \frac{kx^2 - k(x+h)^2}{h\cdot x^2\cdot(x+h)^2} = \lim_{h\to 0} \frac{kx^2 - kx^2 - 2kxh - kh^2}{h\cdot x^2\cdot(x+h)^2}$$

$$= \lim_{h\to 0} \frac{-2kxh - kh^2}{hx^2\cdot(x+h)^2} = \lim_{h\to 0} \frac{-2kx - kh}{x^2(x+h)^2} = \frac{-2k}{x^3}$$

A3 - Derivative rules (inspired by past final worksheet problem 3 and Brightspace problem 2)

Consider the graph of a function $f(x)$ shown to the right. Use this graph to answer each of the following questions.



(a) Let $g(x)$ be the function $g(x) = \dfrac{f(\sin(-3x))}{-2e^{-x}}$.
Find $g'(0)$.

$$\frac{-2e^{-x}\cdot f'(\sin(-3x))\cdot \cos(-3x)\cdot -3 - f(\sin(-3x))\cdot 2e^{-x}}{(-2e^{-x})^2}$$

@ $x=0$:

$$\frac{-2e^{0}\cdot f'(\sin(-3\cdot 0))\cdot \cos(-3\cdot 0)\cdot -3 - f(\sin(-3\cdot 0))\cdot 2e^{0}}{(-2e^{0})^2}$$

$$\frac{-2\cdot f'(0)\cdot 1\cdot -3 - f(0)\cdot 2}{4} = \frac{-2\cdot -\frac{1}{3}\cdot -3 - 3\cdot 2}{4}$$

$$= \frac{-2-6}{4} = \boxed{-2}$$

(b) Let $h(x) = x^{f(x)}$. Find $h'(e^2)$. It may be helpful to know that $e^2 \approx 7.4$

$$x^{f(x)} = e^{f(x)\cdot \ln(x)}$$

$$\frac{d}{dx}\left(e^{f(x)\cdot\ln(x)}\right) = e^{f(x)\cdot\ln(x)}\cdot\left(f'(x)\cdot\ln(x) + f(x)\cdot\frac{1}{x}\right)$$

@ $x=e^2$: $\quad e^{f(e^2)\cdot\ln(e^2)}\cdot\left(f'(e^2)\cdot\ln(e^2) + f(e^2)\cdot\frac{1}{e^2}\right)$

$$= e^{2\cdot 2}\cdot\left(0\cdot 2 + \frac{2}{e^2}\right) = \boxed{2e^2}$$

A4 - Implicit curves and linearization (inspired by part 1 of the class notes and Brightspace problem 10)

(a) Consider the implicit curve given by the equation

$$x^2 + xy + y^2 = 13$$

This curve defines an implicit function $f(x)$ such that $f(4) = -1$ and another implicit function $g(x)$ such that $g(4) = -3$. Find both $f'(4)$ and $g'(4)$.

$$2x + \left(x \cdot \frac{dy}{dx} + 1 \cdot y\right) + 2y \cdot \frac{dy}{dx} = 0$$

$$x \cdot \frac{dy}{dx} + 2y\frac{dy}{dx} = -y - 2x$$

$$\frac{dy}{dx} = \frac{-y-2x}{x+2y}$$

$$f'(4) = \frac{-(-1) - 2\cdot 4}{4 + 2\cdot(-1)} = \frac{1-8}{4-2} = \boxed{\frac{-7}{2}}$$

$$g'(4) = \frac{-(-3) - 2\cdot 4}{4 + 2\cdot(-3)} = \frac{3-8}{4-6} = \boxed{\frac{5}{2}}$$

(b) The number $\frac{4}{(1.9)^2}$ can be approximated using the linearization of the function $f(x) = \frac{4}{x^2}$ at $x = 2$. Find this approximation.

Approximate at $a = 2$

$$f'(x) = \frac{-8}{x^3}$$

$$f'(2) = \frac{-8}{2^3} = -1$$

$$f(2) = \frac{4}{2^2} = 1$$

$$L(x) = f'(2)(x-2) + f(2) = -1(x-2) + 1 = -x + 3$$

$$\frac{4}{(1.9)^2} = f(1.9) \approx L(1.9) = -1.9 + 3 = \boxed{1.1}$$

A5 - Related rates (inspired by past final worksheet problem 5 and Brightspace problem 10)

(a) At what rate is the volume of a cylinder changing if its radius is 1 cm and decreasing at a rate of 2 cm per second, and its height is 3 cm and increasing at a rate of 12 cm per second? Include units in your answer. The volume of a cylinder of radius r and height h is $V = \pi r^2 h$.

$$V = \pi r^2 h$$

$$\frac{dV}{dt} = \pi \cdot \left(r^2 \cdot \frac{dh}{dt} + 2r \cdot \frac{dr}{dt} h \right)$$

$$= \pi \cdot \left(1^2 \cdot 12 + 2 \cdot 1 \cdot (-2) \cdot 3 \right) = 0 \ \text{cm}^3/\text{s}$$

(b) If two resistors with resistances R_1 and R_2 are connected in parallel, then the total resistance R, measured in ohms (Ω) is given by

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

If R_1 and R_2 are increasing at rates of 7 Ω/s and 5 Ω/s respectively, how fast is R changing when $R_1 = 2\ \Omega$ and $R_2 = 1\ \Omega$? Include units in your answer.

$$\frac{1}{R} = \frac{1}{2} + \frac{1}{1} \rightarrow R = \frac{2}{3}$$

$$-R^{-2} \cdot \frac{dR}{dt} = -R_1^{-2} \cdot \frac{dR_1}{dt} - R_2^{-2} \cdot \frac{dR_2}{dt}$$

$$\frac{dR}{dt} = R^2 \cdot \left(R_1^{-2} \cdot \frac{dR_1}{dt} + R_2^{-2} \frac{dR_2}{dt} \right)$$

$$= \left(\frac{2}{3}\right)^2 \cdot \left(\frac{1}{2^2} \cdot 7 + \frac{1}{1^2} \cdot 5 \right) = 3 \ \Omega/\text{s}$$